



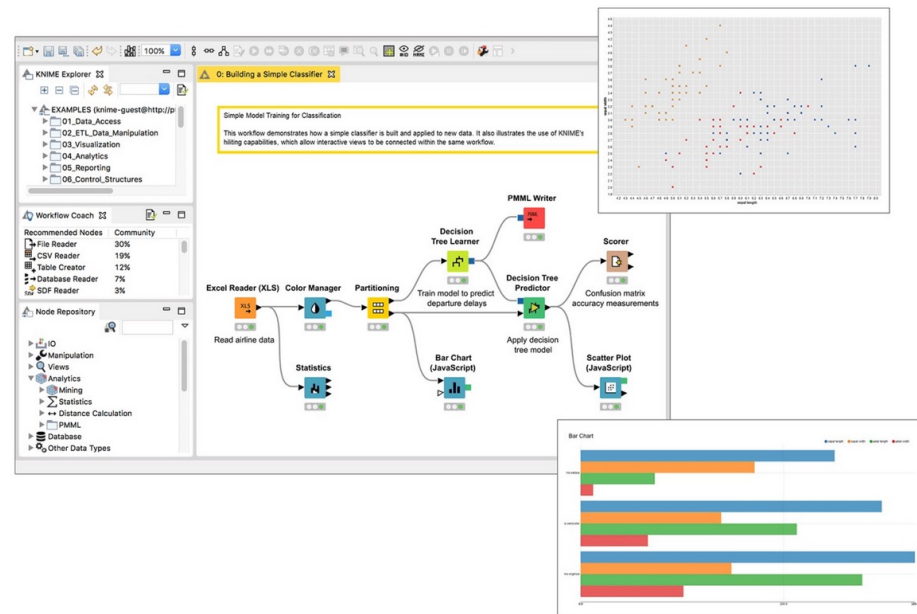
Data Analytics & Data Engineering

เครื่องมือวิทยาการข้อมูล: KNIME

Damrongrit Setsirichok
Waranyu Wongseeree

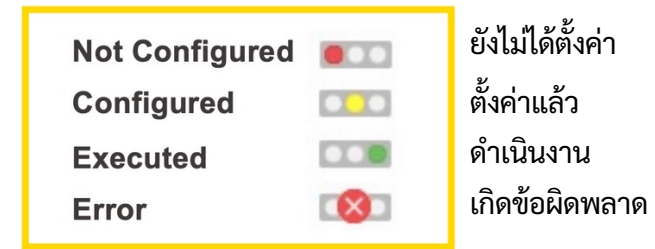
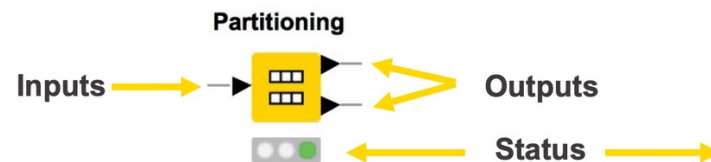
KNIME คืออะไร

- เครื่องมือสร้างกระบวนการจัดการข้อมูล การไหลของข้อมูล การวิเคราะห์ข้อมูล การแสดงภาพและรายงาน
- Open-source สำหรับสร้าง Workflows ข้อมูลในรูปแบบภาพ ช่วยให้ประมวลผล และวิเคราะห์ข้อมูลได้

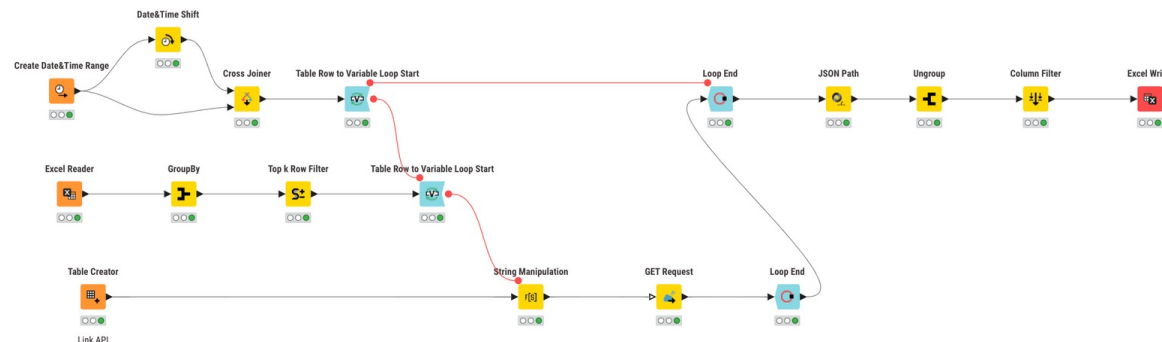


Visual KNIME Workflows

Nodes ดำเนินงานบนข้อมูล

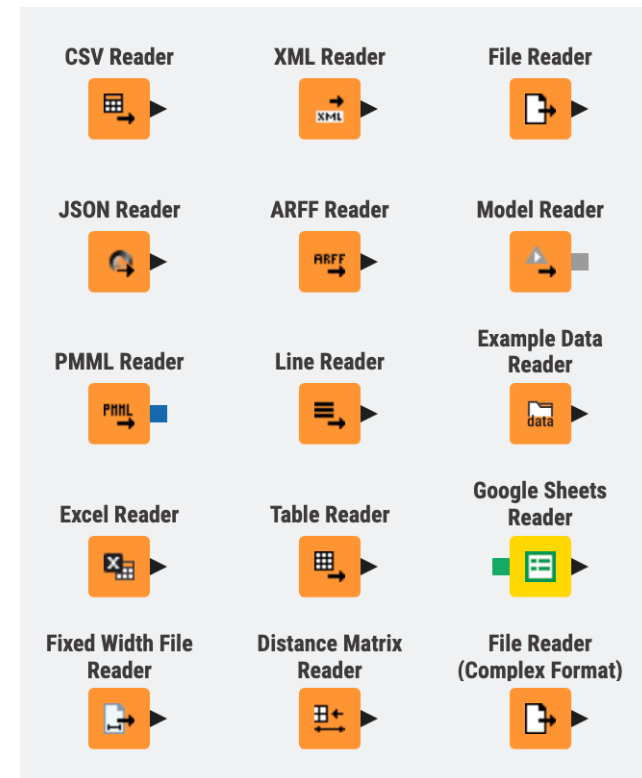


นำ Nodes หลายตัวมารวมกันเพื่อทำ WORKFLOWS



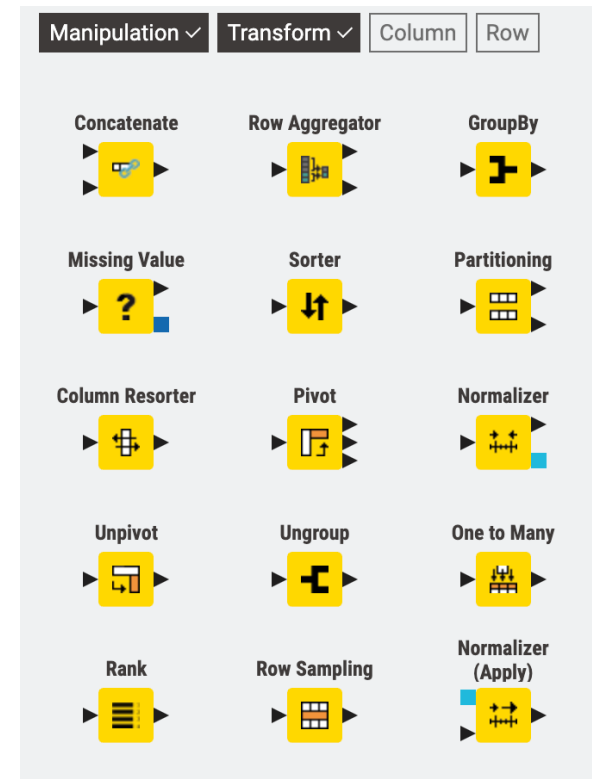
Data Access

- **Databases**
 - MySQL, PostgreSQL, Oracle, etc.
- **Files**
 - CSV, Word, Excel, PDF
 - XML, JSON
 - Images, texts
- **Other**
 - TwitterAPI, GoogleAPI
 - REST API
 - Amazon S3, Azure Blob Store



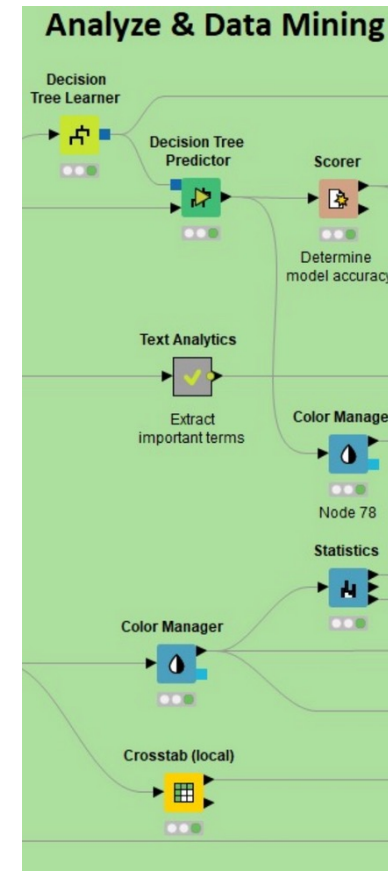
Transformation

- **Preprocessing (การประมวลผลเบื้องต้น)**
 - Row, Column, matrix based
- **Data blending (การนำข้อมูลมารวมกัน)**
 - Join, Concatenate, Append
- **Aggregation (การรวมกลุ่ม)**
 - Grouping, Pivoting, Binning



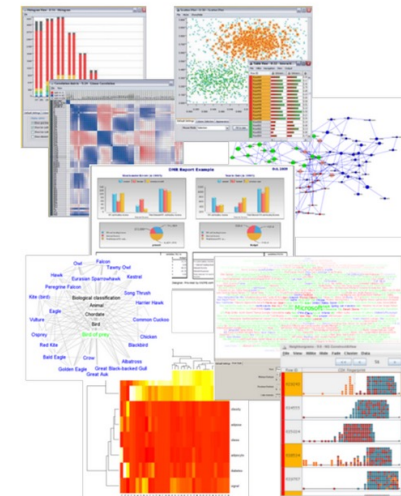
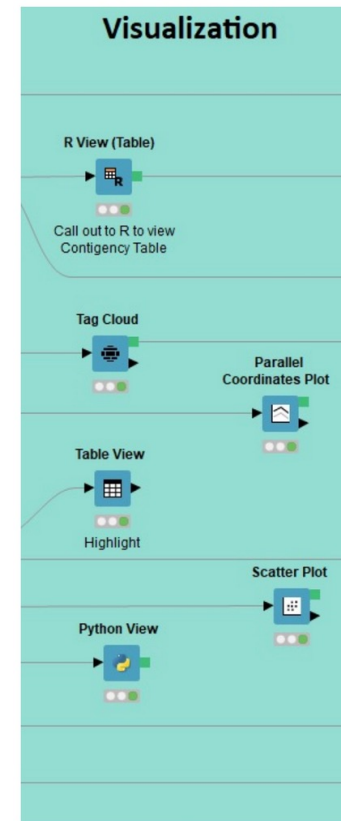
Analysis & Data Mining

- **Regression**
 - Linear, Logistic
- **Classification**
 - Decision tree, ensembles, SVM, MLP
- **Clustering**
 - k-means, DBSCAN, hierarchical
- **Validation**
 - Cross-validation, scoring
- **Deep Learning**
 - Keras, DL4J
- **External**
 - R, Python, H2O



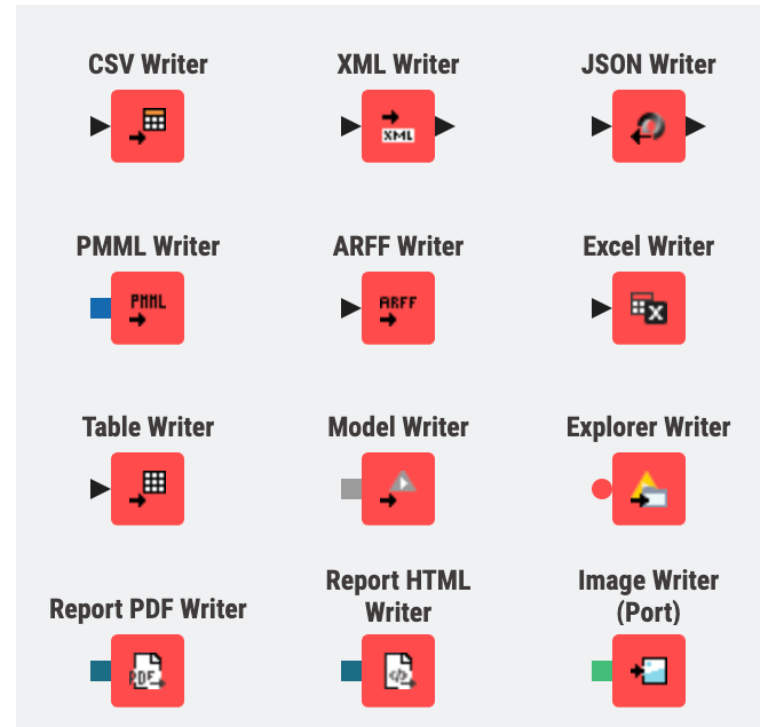
Visualization

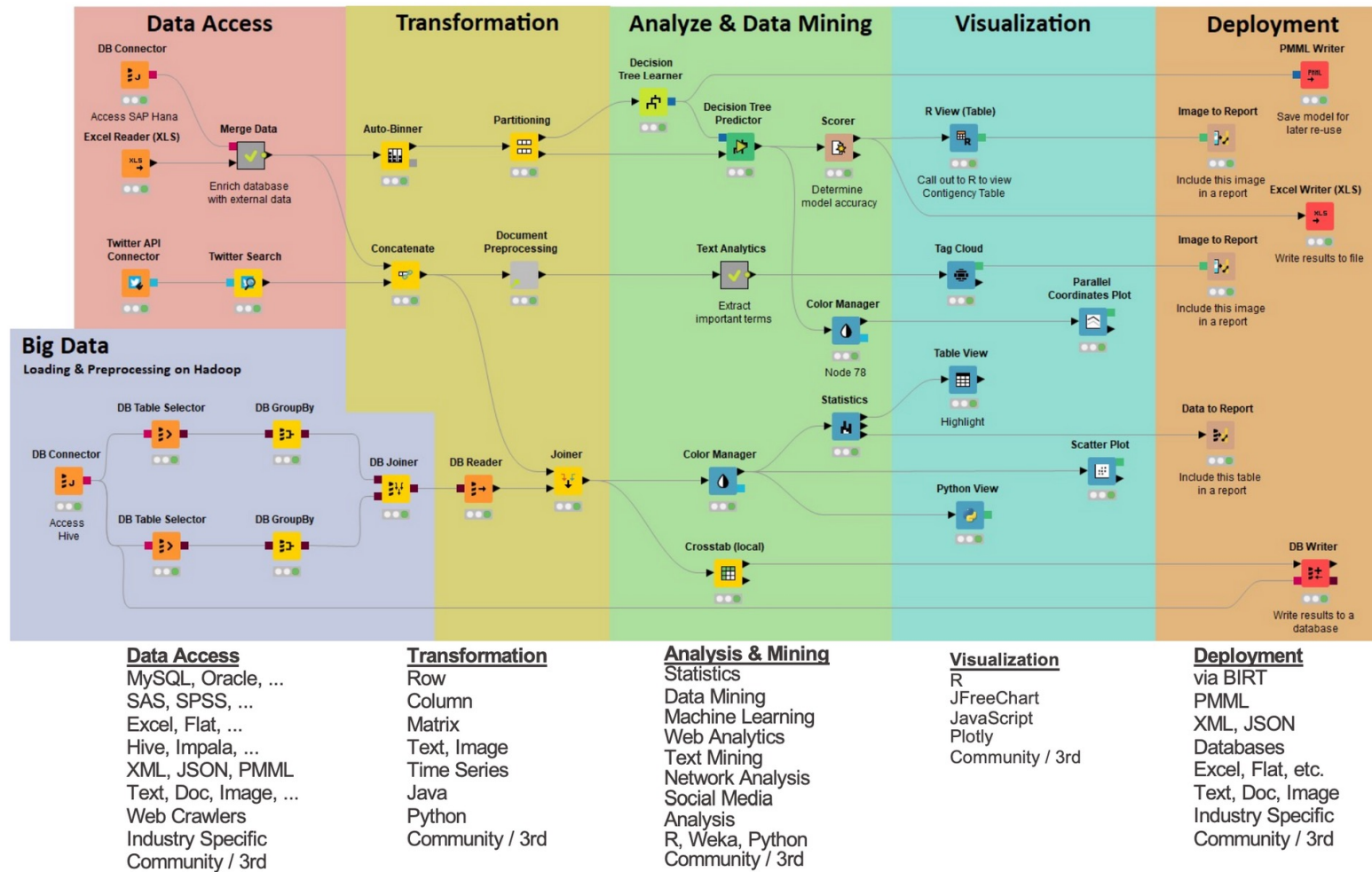
- **Interactive Visualization**
- **JavaScript-based Nodes**
 - Scatter Plot, Box Plot, Line Plot
 - Decision Tree
- **Script-based Visualizations**
 - R, Python



Deployment

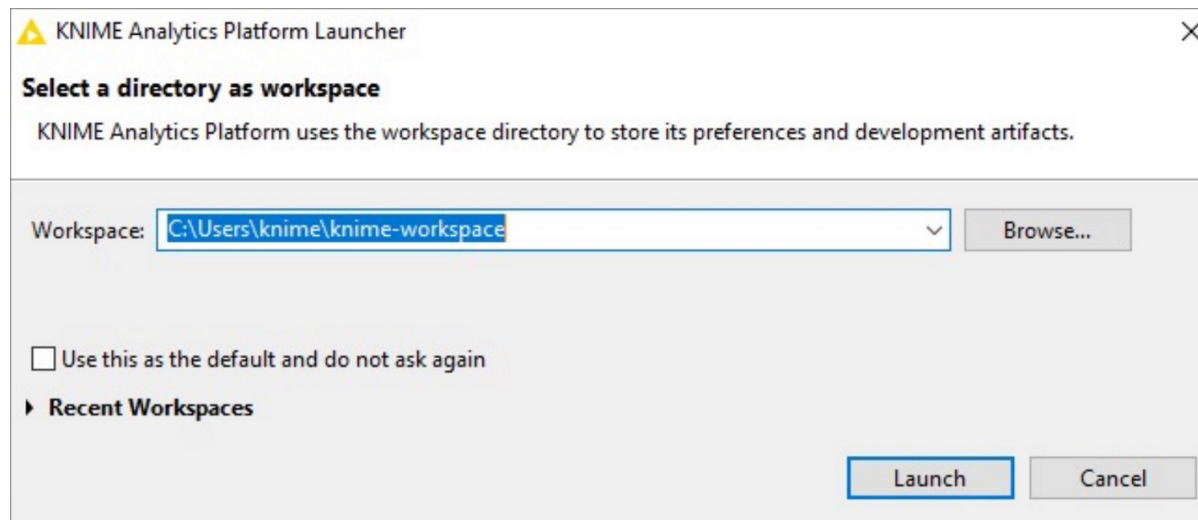
- Database
- Files
 - Excel, CSV
 - XML
 - PMML
 - Amazon S3, Azure Blob Store





KNIME Workspace

- Workspace เป็น Folder/Directory ที่ใช้เก็บ Workflows



KNIME Analytics Platform Workbench

The screenshot displays the KNIME Analytics Platform Workbench interface. The main workspace shows a workflow titled "My first Workflow" with four nodes: File Reader, Row Filter, Column Filter, and Table Writer. The Row Filter node is highlighted with a yellow box, and its configuration dialog is open on the right, showing the "Row Filter" node description and "Dialog Options".

Labels pointing to specific interface elements include:

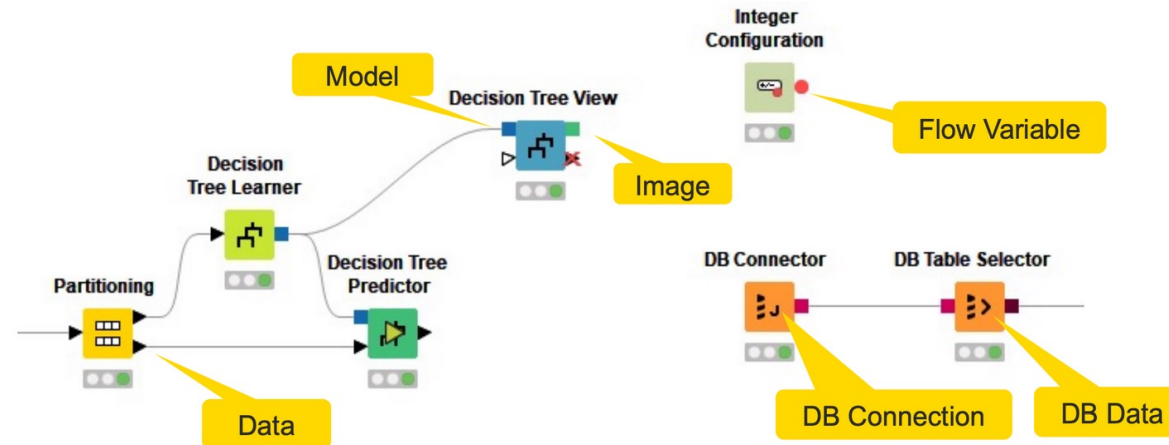
- KNIME Explorer**: Points to the left sidebar showing the project structure.
- Workflow Coach**: Points to the "Recommended Nodes" list in the left sidebar.
- Node Repository**: Points to the "Node Repository" section in the left sidebar.
- Workflow Editor**: Points to the central workspace where the workflow is built.
- Node Description**: Points to the "Row Filter" node description in the right sidebar.
- Console & Node Monitor**: Points to the bottom right panel showing the execution status and data output of the selected node.
- Outline**: Points to the "Outline" panel in the bottom left corner.

The "Console & Node Monitor" panel displays the following data output for the "Row Filter" node:

ID	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex	capital-gain	capital-loss	hours
Row0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male	2174	0	40
Row1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	13
Row2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0	40

Inserting and Connecting Nodes

- การนำเข้า Nodes มาใน Workspace ทำได้โดยการลาก Node ที่ต้องการ นำมาวางที่ Workspace
- เชื่อม Node โดยการคลิกซ้ายแล้วลากเชื่อม Node A -> Node B



Node State

- ในแต่ละ Node จะมีอยู่ทั้งหมด 4 States

File Reader



ยังไม่ได้ตั้งค่า (Not Configured)
Node กำลังรอตั้งค่าหรือนำเข้าข้อมูล

File Reader



ตั้งค่าแล้ว (Configured)
Node ตั้งค่าเรียบร้อยแล้ว พร้อมที่จะดำเนินงาน (Executed)

File Reader



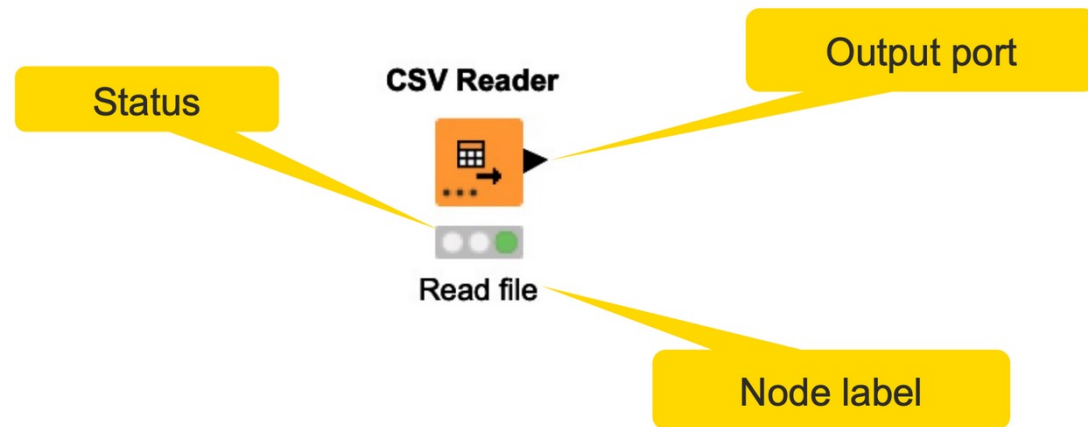
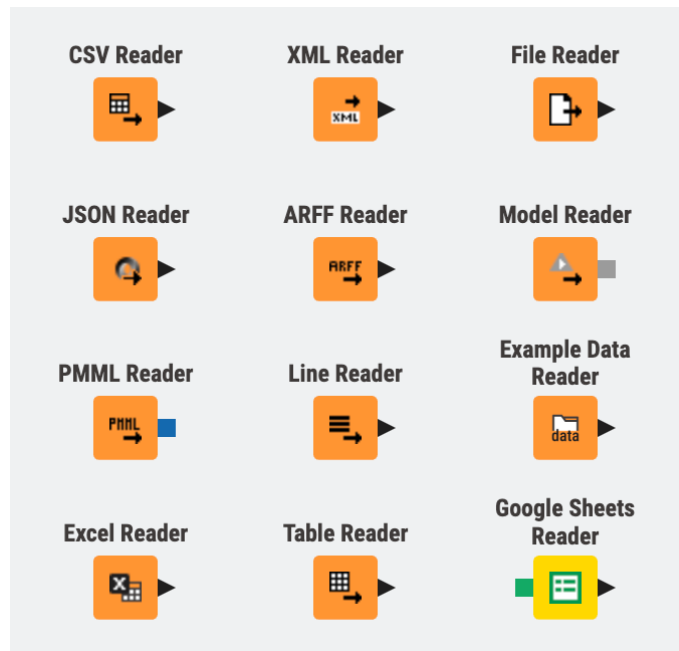
ดำเนินงานแล้ว (Executed)
Node ดำเนินงานเรียบร้อยแล้ว ซึ่งจะแสดงผลลัพธ์ที่ Console & Node Monitor

File Reader



เกิดข้อผิดพลาด (Error)
Node ที่มีข้อผิดพลาดเกิดขึ้นระหว่างดำเนินงาน (Executed)

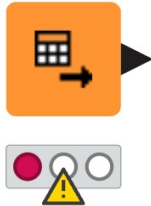
Importing Data Accessing Files



CSV Reader

File Path

CSV Reader



Preview Imported Data

Dialog - 4:2 - CSV Reader (Customer)

Settings Transformation Advanced Settings Limit Rows Encoding Flow Variables Job Manager Selection Memory Policy

Input location

Read from Local File System

Mode ☒ File ☐ Files in folder

File Customers.csv Browse...

Reader options

Format

Autodetect format

Column delimiter Row delimiter ☒ Line break ☐ Custom \n

Quote char Quote escape char

Comment char

☒ Has column header ☐ Has RowID

☐ Support short data rows ☐ Prepend file index to RowID

Preview

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	I	Custo...	S Name	S Email	S Phone	S City
Row0	1	Suwannee	suwannee1@hotmail.com	+66-836666371	Udon Thani	
Row1	2	Chaiwat	chaiwat2@gmail.com	+66-851180162	Nakhon R...	
Row2	3	Siriwan	siriwan3@gmail.com	+66-884670607	Khon Kaen	
Row3	4	Wanchai	wanchai4@hotmail.com	+66-855634814	Phuket	
Row4	5	Somsri	somsri5@yahoo.com	+66-815221506	Nakhon Si...	
Row5	6	Wanchai	wanchai6@yahoo.com	+66-853839445	Bangkok	
Row6	7	Suthida	suthida7@yahoo.com	+66-879708486	Udon Thani	
Row7	8	Suwannee	suwannee8@gmail.com	+66-886870853	Bangkok	
Row8	9	Nattapong	nattapong9@hotmail.com	+66-829488453	Khon Kaen	
Row9	10	Pimchanok	pimchanok10@gmail.com	+66-843782099	Surat Thani	
Row10	11	Yingluck	yingluck11@gmail.com	+66-858374782	Udon Thani	
Row11	12	Nattaya	nattaya12@yahoo.com	+66-875229886	Udon Thani	
Row12	13	Nattaya	nattaya13@yahoo.com	+66-822382506	Pattaya	
Row13	14	Nattapong	nattapong14@yahoo.com	+66-885729663	Hat Yai	
Row14	15	Suwannee	suwannee15@yahoo.com	+66-838496160	Phuket	

OK Apply Cancel

1. Browse .csv file

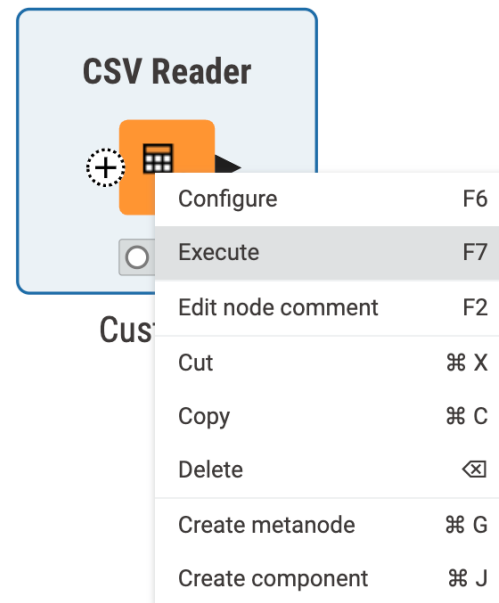
2. Click OK

CSV Reader

CSV Reader



สถานะจะเปลี่ยนเป็นตั้งค่าเรียบร้อยแล้วพร้อม Execute



CSV Reader

CSV Reader

Configure

Open output port

Reset

Edit node comment

Cut

Copy

Delete

Create metanode

Create component

F6

F8

F2

⌘ X

⌘ C

⌘

⌘ G

⌘ J

Variable Output

Flow variables

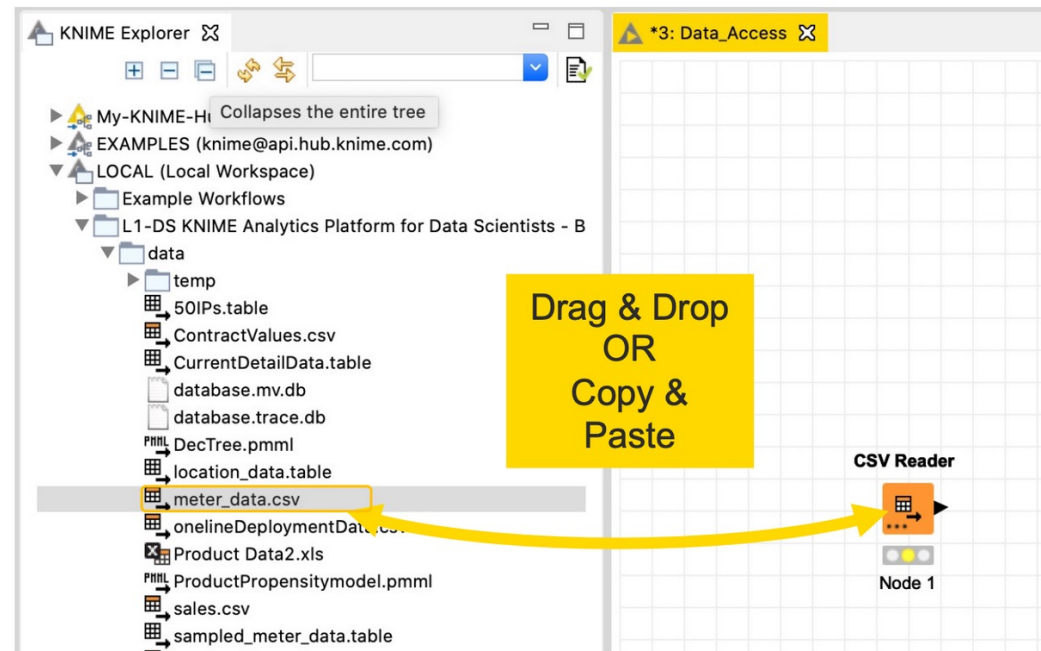
1: File Table

Table

Statistics

File Table (Table)						
Rows: 200 Columns: 5						
#	RowID	CustomerID	Name	Email	Phone	City
		Number (integer)	String	String	String	String
☐	1	Row0	1	Suwannee	suwannee1@hotmail.com	+66-836666371
☐	2	Row1	2	Chaiwat	chaiwat2@gmail.com	+66-851180162
☐	3	Row2	3	Siriwan	siriwan3@gmail.com	+66-884670607
☐	4	Row3	4	Wanchai	wanchai4@hotmail.com	+66-855634814
☐	5	Row4	5	Somsri	somsri5@yahoo.com	+66-815221506
☐	6	Row5	6	Wanchai	wanchai6@yahoo.com	+66-853839445
☐	7	Row6	7	Suthida	suthida7@yahoo.com	+66-879708486
☐	8	Row7	8	Suwannee	suwannee8@gmail.com	+66-886870853
☐	9	Row8	9	Nattapong	nattapong9@hotmail.com	+66-829488453
☐	10	Row9	10	Pimchanok	pimchanok10@gmail.com	+66-843782099
☐	11	Row...	11	Yingluck	yingluck11@gmail.com	+66-858374782
☐	12	Row...	12	Nattaya	nattaya12@yahoo.com	+66-875229886
☐	13	Row...	13	Nattaya	nattaya13@yahoo.com	+66-822382506
☐	14	Row...	14	Nattapong	nattapong14@yahoo.com	+66-885729663
☐	15	Row...	15	Suwannee	suwannee15@yahoo.com	+66-838496160
☐	16	Row...	16	Arthit	arthit16@hotmail.com	+66-843878036
☐	17	Row...	17	Suthida	suthida17@gmail.com	+66-847036318
☐	18	Row...	18	Chaiwat	chaiwat18@gmail.com	+66-866762690
☐	19	Row...	19	Preecha	preecha19@hotmail.com	+66-811405226
☐	20	Row...	20	Somchai	somchai20@gmail.com	+66-876791657
☐	21	Row...	21	Sasithorn	sasithorn21@hotmail.com	+66-849604058
☐	22	Row...	22	Nattaya	nattaya22@hotmail.com	+66-859738646
☐	23	Row...	23	Prayut	prayut23@gmail.com	+66-821401230
☐	24	Row...	24	Nattaya	nattaya24@yahoo.com	+66-857252705
☐	25	Row...	25	Somchai	somchai25@yahoo.com	+66-829822117
☐	26	Row...	26	Anucha	anucha26@hotmail.com	+66-849536832

CSV Reader



Excel Reader

File Path

Excel Reader



Preview Imported Data

Dialog - 4:1 - Excel Reader (Orders)

File and Sheet | Data Area | Advanced | Transformation | Flow Variables | Job Manager Selection | Memory Policy

Input Location

Read from: Local File System

Mode: ☒ File ☐ Files in folder

File: Orders.xlsx Browse...

Select Sheet

☒ First with data (Sheet1)

☐ By name: Sheet1

☐ By position: 0 (Position starts with 0.)

Preview File Content

Preview with current settings

The suggested column types are based on the first 10000 rows only. See 'Advanced Settings' tab.

Row ID	OrderID	Custo...	Product	Quantity	OrderD...
Row0	1	52	Smartwatch	1	2024-01-32
Row1	2	93	Smartphone	2	2025-10-60
Row2	3	15	Tablet	1	2025-01-01
Row3	4	72	Smartwatch	1	2023-10-09
Row4	5	61	Smartwatch	3	2023/09/08
Row5	6	21	Headphon...	2	2023/08/17
Row6	7	83	Smartphone	4	2023/05/23
Row7	8	87	Smartwatch	4	2023/04/15
Row8	9	75	Smartphone	3	2024/11/23
Row9	10	75	Tablet	3	2024-07-12
Row10	11	88	Headphon...	2	2023/03/31
Row11	12	100	Tablet	4	2024-08-27
Row12	13	24	Headphon...	4	2024/03/08
Row13	14	3	Smartwatch	4	2023/02/02
Row14	15	22	Headphon...	1	2023-01-31
Row15	16	53	Laptop	4	2023-04-06
Row16	17	2	Smartphone	1	2023/08/12
Row17	18	88	Smartphone	2	2023-08-27
Row18	19	30	Laptop	1	2024/06/01
Row19	20	38	Tablet	2	2024/09/08
Row20	21	2	Laptop	4	2023/01/28
Row21	22	64	Smartwatch	4	2024-07-28

OK Apply Cancel ?

1. Browse .csv file

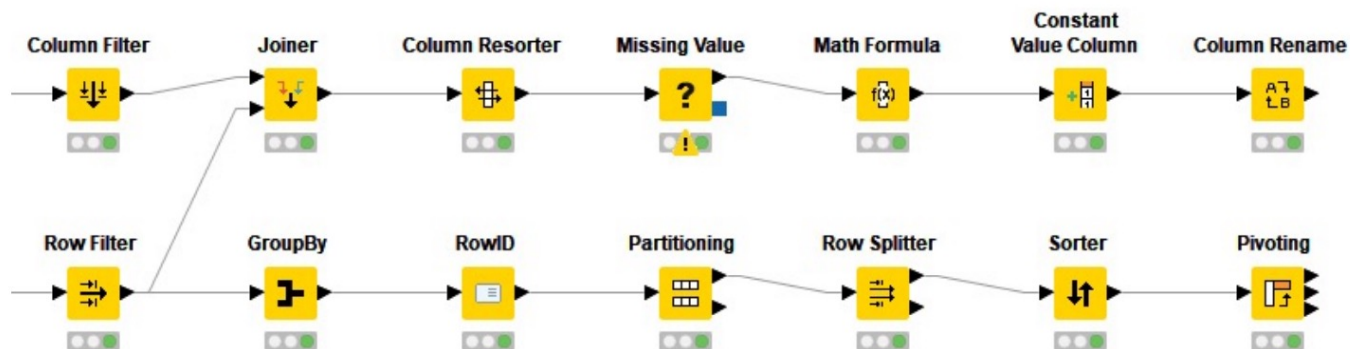
2. Click OK

Excel Reader

File Table (Table)								
Rows: 210			Columns: 5					
☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String	☐
☐	1	Row0	1	52	Smartwatch	1	2024-01-32	
☐	2	Row1	2	93	Smartphone	2	2025-10-60	
☐	3	Row2	3	15	Tablet	1	2025-01-01	
☐	4	Row3	4	72	Smartwatch	1	2023-10-09	
☐	5	Row4	5	61	Smartwatch	3	2023/09/08	
☐	6	Row5	6	21	Headphones	2	2023/08/17	
☐	7	Row6	7	83	Smartphone	4	2023/05/23	
☐	8	Row7	8	87	Smartwatch	4	2023/04/15	
☐	9	Row8	9	75	Smartphone	3	2024/11/23	
☐	10	Row9	10	75	Tablet	3	2024-07-12	
☐	11	Row...	11	88	Headphones	2	2023/03/31	
☐	12	Row...	12	100	Tablet	4	2024-08-27	
☐	13	Row...	13	24	Headphones	4	2024/03/08	
☐	14	Row...	14	3	Smartwatch	4	2023/02/02	
☐	15	Row...	15	22	Headphones	1	2023-01-31	
☐	16	Row...	16	53	Laptop	4	2023-04-06	
☐	17	Row...	17	2	Smartphone	1	2023/08/12	
☐	18	Row...	18	88	Smartphone	2	2023-08-27	
☐	19	Row...	19	30	Laptop	1	2024/06/01	
☐	20	Row...	20	38	Tablet	2	2024/09/08	
☐	21	Row...	21	2	Laptop	4	2023/01/28	
☐	22	Row...	22	64	Smartwatch	4	2024-07-28	
☐	23	Row...	23	60	Smartwatch	2	2023-07-23	
☐	24	Row...	24	21	Smartphone	3	2024/10/27	
☐	25	Row...	25	33	Smartwatch	2	2024/12/19	
☐	26	Row...	26	76	Smartphone	3	2024/07/12	

Data Manipulation Node

- Data Manipulation จะเป็น Node สีเหลือง
- ใช้ในการแปลง (Transform) ข้อมูล



String Replacer

File Table (Table)

Rows: 210 | Columns: 5

☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String
☐	1	Row0	1	52	Smartwatch	1	2024-01-32
☐	2	Row1	2	93	Smartphone	2	2025-10-60
☐	3	Row2	3	15	Tablet	1	2025-01-01
☐	4	Row3	4	72	Smartwatch	1	2023-10-09
☐	5	Row4	5	61	Smartwatch	3	2023/09/08
☐	6	Row5	6	21	Headphones	2	2023/08/17
☐	7	Row6	7	83	Smartphone	4	2023/05/23

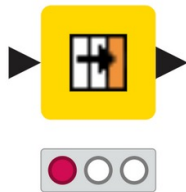
Input with replaced strings (Table)

Rows: 210 | Columns: 5

☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String
☐	1	Row0	1	52	Smartwatch	1	2024-01-32
☐	2	Row1	2	93	Smartphone	2	2025-10-60
☐	3	Row2	3	15	Tablet	1	2025-01-01
☐	4	Row3	4	72	Smartwatch	1	2023-10-09
☐	5	Row4	5	61	Smartwatch	3	2023-09-08
☐	6	Row5	6	21	Headphones	2	2023-08-17
☐	7	Row6	7	83	Smartphone	4	2023-05-23

String Replacer

String Replacer



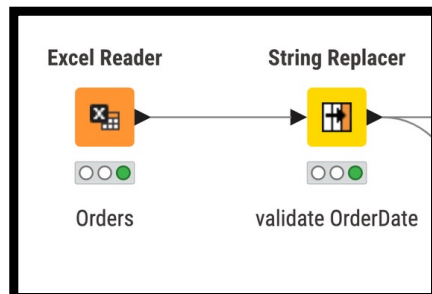
ระบุ Target Column



ระบุ Pattern



ระบุ Replacement Text



Dialog - 4:4 - String Replacer (validate OrderDate)

Column Selection

Target column
OrderDate

Find & Replace

Pattern type
Literal Wildcard Regular expression

Case sensitive
Case sensitive Case insensitive

Pattern
/

Replacement text
-

Replacement strategy
Whole string All occurrences

Output

☐ Append new column

Cancel Ok and Execute

เลือก All Occurences

กด Ok and Execute

Cell Splitter

Input with replaced strings (Table)

Rows: 210 | Columns: 5

#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String
1	Row0	1	52	Smartwatch	1	2024-01-32
2	Row1	2	93	Smartphone	2	2025-10-60
3	Row2	3	15	Tablet	1	2025-01-01
4	Row3	4	72	Smartwatch	1	2023-10-09
5	Row4	5	61	Smartwatch	3	2023-09-08
6	Row5	6	21	Headphones	2	2023-08-17
7	Row6	7	83	Smartphone	4	2023-05-23

Output Table (Table)

Rows: 210 | Columns: 8

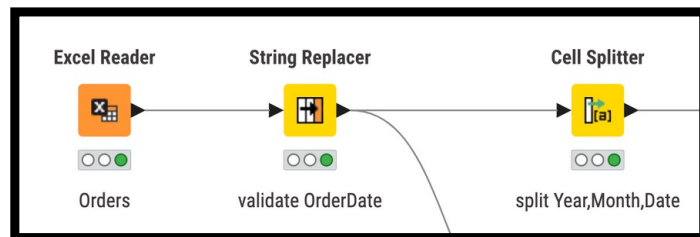
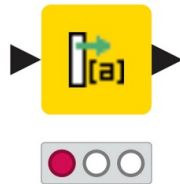
#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String	OrderDate_Arr[0] Number (integer)	OrderDate_Arr[1] Number (integer)	OrderDate_Arr[2] Number (integer)
1	Row0	1	52	Smart...	1	2024-01-32	2024	1	32
2	Row1	2	93	Smart...	2	2025-10-60	2025	10	60
3	Row2	3	15	Tablet	1	2025-01-01	2025	1	1
4	Row3	4	72	Smart...	1	2023-10-09	2023	10	9
5	Row4	5	61	Smart...	3	2023-09-08	2023	9	8
6	Row5	6	21	Head...	2	2023-08-17	2023	8	17
7	Row6	7	83	Smart...	4	2023-05-23	2023	5	23

Cell Splitter

เลือก Column ที่ต้องการ Split →

ระบุตัวแบ่ง (Delimiter) →

Cell Splitter



Dialog - 4:5 - Cell Splitter (split Year,Month,Date)

Settings | Flow Variables | Job Manager Selection | Memory Policy

Column to split

Select a column: ☐ Remove input column

Settings

Enter a delimiter: ☐ Use \ as escape character

Enter a quotation character: (leave empty for none.)

☒ Remove leading and trailing white space chars (trim)

Output

☐ As list ☐ As set (remove duplicates) ☒ As new columns

☐ Split input column name for output column names

☐ Set array size

☒ Guess size and column types (requires additional data table scan)

☐ Scan limit (number of lines to guess on)

Missing Value Handling

☐ Create empty string cells instead of missing string cells

OK →

Column Renamer

Output Table (Table)										
Rows: 210		Columns: 8								
☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String	OrderDate_Arr[0] Number (integer)	OrderDate_Arr[1] Number (integer)	OrderDate_Arr[2] Number (integer)
☐	1	Row0	1	52	Smartwatch	1	2024-01-32	2024	1	32
☐	2	Row1	2	93	Smartwatch	2	2025-10-60	2025	10	60
☐	3	Row2	3	15	Tablet	1	2025-01-01	2025	1	1
☐	4	Row3	4	72	Smartwatch	1	2023-10-09	2023	10	9
☐	5	Row4	5	61	Smartwatch	3	2023-09-08	2023	9	8
☐	6	Row5	6	21	Headphones	2	2023-08-17	2023	8	17
☐	7	Row6	7	83	Smartphone	4	2023-05-23	2023	5	23

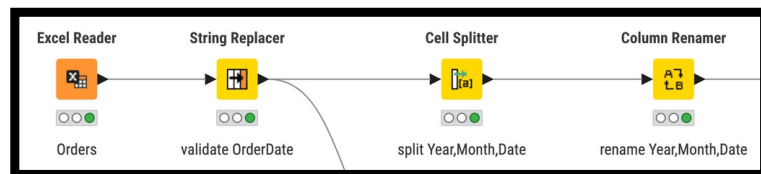
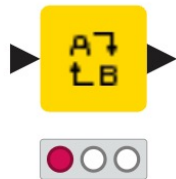
↓

Included Rows (Table)										
Rows: 208		Columns: 8								
☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String	Year Number (integer)	Month Number (integer)	Date Number (integer)
☐	1	Row2	3	15	Tablet	1	2025-01-01	2025	1	1
☐	2	Row3	4	72	Smartwatch	1	2023-10-09	2023	10	9
☐	3	Row4	5	61	Smartwatch	3	2023-09-08	2023	9	8
☐	4	Row5	6	21	Headphones	2	2023-08-17	2023	8	17
☐	5	Row6	7	83	Smartphone	4	2023-05-23	2023	5	23

Column Renamer

เลือก Column ที่ต้องการเปลี่ยนชื่อ →

Column Renamer



Dialog - 4:6 - Column Renamer (rename Year,Month,Date)

Column	New name	
OrderDate_Arr[0]	Year	🗑️
OrderDate_Arr[1]	Month	🗑️
OrderDate_Arr[2]	Date	🗑️

⊕ Add column

Cancel Ok and Execute

Row Filter

Output Table (Table)

Rows: 210 | Columns: 8

☐	#	RowID	OrderID Number (integ...)	CustomerID Number (integ...)	Product String	Quantity Number (integ...)	OrderDate String	Year Number (integ...)	Month Number (integ...)	Date Number (integ...)	
☐	1	Row0	1	52	Smartwatch	1	2024-01-32	2024	1	32	
☐	2	Row1	2	93	Smartphone	2	2025-10-60	2025	10	60	
☐	3	Row2	3	15	Tablet	1	2025-01-01	2025	1	1	
☐	4	Row3	4	72	Smartwatch	1	2023-10-09	2023	10	9	
☐	5	Row4	5	61	Smartwatch	3	2023-09-08	2023	9	8	
☐	6	Row5	6	21	Headphones	2	2023-08-17	2023	8	17	
☐	7	Row6	7	83	Smartphone	4	2023-05-23	2023	5	23	

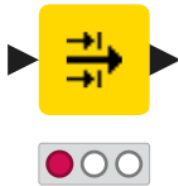
Included Rows (Table)

Rows: 208 | Columns: 8

☐	#	RowID	OrderID Number (integ...)	CustomerID Number (integ...)	Product String	Quantity Number (integ...)	OrderDate String	Year Number (integ...)	Month Number (integ...)	Date Number (integ...)	
☐	1	Row2	3	15	Tablet	1	2025-01-01	2025	1	1	
☐	2	Row3	4	72	Smartwatch	1	2023-10-09	2023	10	9	
☐	3	Row4	5	61	Smartwatch	3	2023-09-08	2023	9	8	
☐	4	Row5	6	21	Headphones	2	2023-08-17	2023	8	17	
☐	5	Row6	7	83	Smartphone	4	2023-05-23	2023	5	23	

Row Filter

Row Filter



Dialog - 4:45 - Row Filter (filter not valid date)

Filter

Criterion 1 ↑ ↓ 🗑️

Filter column: ▼ Operator: ▼

Value: ^ ▼

⊕ Add criterion

Output เพิ่มเงื่อนไขในการ Filter

Column domains: Retain Compute

Filter behavior: Output matching rows Output non-matching rows

Cancel Ok and Execute

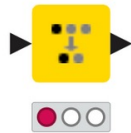
Duplicate Row Filter

Included Rows (Table)										
Rows: 208 Columns: 8										
☐	#	RowID	OrderID Number (integ...	CustomerID Number (integ...	Product String	Quantity Number (integ...	OrderDate String	Year Number (integ...	Month Number (integ...	Date Number (integ...
☐	14	Row...	16	53	Laptop	4	2023-04-06	2023	4	6
☐	29	Row...	31	91	Laptop	1	2023-10-12	2023	10	12
☐	200	Row...	16	53	Laptop	4	2023-04-06	2023	4	6
☐	201	Row...	31	91	Laptop	1	2023-10-12	2023	10	12

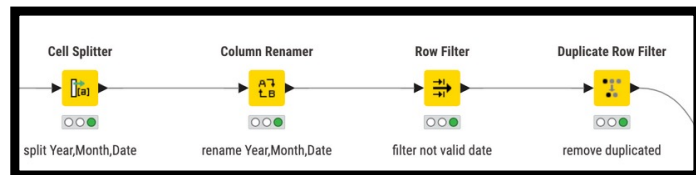
Filtered/Labeled Data (Table)										
Rows: 198 Columns: 8										
☐	#	RowID	OrderID Number (integ...	CustomerID Number (integ...	Product String	Quantity Number (integ...	OrderDate String	Year Number (integ...	Month Number (integ...	Date Number (integ...
☐	14	Row...	16	53	Laptop	4	2023-04-06	2023	4	6
☐	29	Row...	31	91	Laptop	1	2023-10-12	2023	10	12

Duplicate Row Filter

Duplicate Row Filter



Filter Column Name ใดบ้าง



Dialog - 4:3 - Duplicate Row Filter (remove duplicated)

Duplicate detection

Choose columns for duplicates detection

Manual Wildcard Regex Type

Search

Excludes

- Year
- Month
- Date

Includes

- OrderID
- CustomerID
- Product
- Quantity
- OrderDate

Any unknown column

Duplicate handling

Duplicate rows

- ☒ Remove duplicate rows
- ☐ Keep duplicate rows

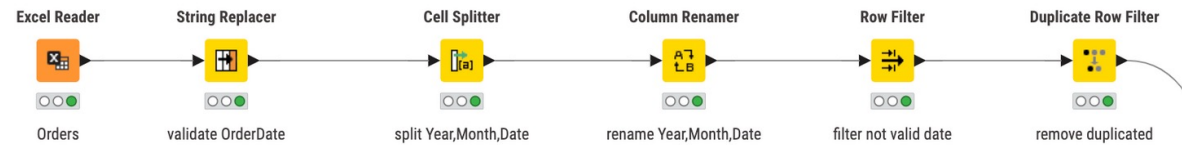
Row chosen in case of duplicate

- ☒ First
- ☐ Last
- ☐ Minimum of
- ☐ Maximum of

Cancel Ok and Execute

ถ้า OrderId, CustomerId, Product, Quantity, และ OrderDate เหมือนกัน
ทุกประการจะถือว่า Duplicate

Workflow



File Table (Table)										
Rows: 210		Columns: 5								
☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate String			
☐	1	Row0	1	52	Smartwatch	1	2024-01-32			
☐	2	Row1	2	93	Smartphone	2	2025-10-60			
☐	3	Row2	3	15	Tablet	1	2025-01-01			
☐	4	Row3	4	72	Smartwatch	1	2023-10-09			
☐	5	Row4	5	61	Smartwatch	3	2023/09/08			
☐	6	Row5	6	21	Headphones	2	2023/08/17			
☐	7	Row6	7	83	Smartphone	4	2023/05/23			

Filtered/Labeled Data (Table)										
Rows: 198		Columns: 8								
☐	#	RowID	OrderID Number (integ...)	CustomerID Number (integ...)	Product String	Quantity Number (integ...)	OrderDate String	Year Number (integ...)	Month Number (integ...)	Date Number (integ...)
☐	1	Row2	3	15	Tablet	1	2025-01-01	2025	1	1
☐	2	Row3	4	72	Smartwatch	1	2023-10-09	2023	10	9
☐	3	Row4	5	61	Smartwatch	3	2023-09-08	2023	9	8
☐	4	Row5	6	21	Headphones	2	2023-08-17	2023	8	17
☐	5	Row6	7	83	Smartphone	4	2023-05-23	2023	5	23

String to Date&Time

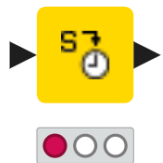
Input with replaced strings (Table)							
Rows: 210 Columns: 5							
☐	#	RowID	OrderID <i>Number (integer)</i>	CustomerID <i>Number (integer)</i>	Product <i>String</i>	Quantity <i>Number (integer)</i>	OrderDate <i>String</i>
☐	1	Row0	1	52	Smartwatch	1	2024-01-32
☐	2	Row1	2	93	Smartphone	2	2025-10-60
☐	3	Row2	3	15	Tablet	1	2025-01-01
☐	4	Row3	4	72	Smartwatch	1	2023-10-09
☐	5	Row4	5	61	Smartwatch	3	2023-09-08
☐	6	Row5	6	21	Headphones	2	2023-08-17
☐	7	Row6	7	83	Smartphone	4	2023-05-23

Output table (Table)							
Rows: 210 Columns: 5							
☐	#	RowID	OrderID <i>Number (integer)</i>	CustomerID <i>Number (integer)</i>	Product <i>String</i>	Quantity <i>Number (integer)</i>	OrderDate <i>Local Date</i>
☐	1	Row0	1	52	Smartwatch	1	?
☐	2	Row1	2	93	Smartphone	2	?
☐	3	Row2	3	15	Tablet	1	2025-01-01
☐	4	Row3	4	72	Smartwatch	1	2023-10-09
☐	5	Row4	5	61	Smartwatch	3	2023-09-08
☐	6	Row5	6	21	Headphones	2	2023-08-17
☐	7	Row6	7	83	Smartphone	4	2023-05-23

String to Date&Time

เลือก Column OrderDate ในการ
แปลง String to Date&Time

String to Date&Time



Dialog - 4:11 - String to Date&Time

Options | Flow Variables | Job Manager Selection | Memory Policy

☒ Manual Selection ☐ Wildcard/Regex Selection

Exclude

Filter

S Product

☐ Enforce exclusion

Include

Filter

S OrderDate

☒ Enforce inclusion

Replace/Append Selection

☐ Append selected columns Suffix of appended columns: (Date&Time)

☒ Replace selected columns

Type and Format Selection

New type: Date Date format: yyyy-MM-dd

Locale: en-US Content of the first cell: 2024-01-32

Guess data type and format

Abort Execution

☐ Fail on error

OK Apply Cancel ?

เลือก Type และกำหนด
Date format

ถ้า Fail ให้ขึ้น Error

Missing Value

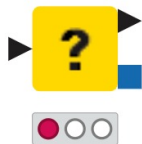
Output table (Table)							
Rows: 210 Columns: 5							
☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate Local Date
☐	1	Row0	1	52	Smartwatch	1	?
☐	2	Row1	2	93	Smartphone	2	?
☐	3	Row2	3	15	Tablet	1	2025-01-01
☐	4	Row3	4	72	Smartwatch	1	2023-10-09
☐	5	Row4	5	61	Smartwatch	3	2023-09-08
☐	6	Row5	6	21	Headphones	2	2023-08-17
☐	7	Row6	7	83	Smartphone	4	2023-05-23

Output table (Table)							
Rows: 208 Columns: 5							
☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate Local Date
☐	1	Row2	3	15	Tablet	1	2025-01-01
☐	2	Row3	4	72	Smartwatch	1	2023-10-09
☐	3	Row4	5	61	Smartwatch	3	2023-09-08
☐	4	Row5	6	21	Headphones	2	2023-08-17
☐	5	Row6	7	83	Smartphone	4	2023-05-23

Missing Value

OrderDate
Local Date
?
?

Missing Value



Dialog - 4:15 - Missing Value

Default Column Settings Flow Variables Job Manager Selection Memory Policy

Number (integer)	Do nothing
String	Do nothing
Local Date	Remove Row*

OK Apply Cancel ?

ระบุกรณีเป็น Number, String, Local Date เมื่อเกิด Missing Value จะให้ทำอะไรต่อไป

Do nothing

✓ Do nothing

Fix Value

Linear Interpolation*

Maximum

Mean

Median

Minimum

Most Frequent Value

Date&Time Part Extractor

Output table (Table)

Rows: 208 | Columns: 5

☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate Local Date
☐	1	Row2	3	15	Tablet	1	2025-01-01
☐	2	Row3	4	72	Smartwatch	1	2023-10-09
☐	3	Row4	5	61	Smartwatch	3	2023-09-08
☐	4	Row5	6	21	Headphones	2	2023-08-17
☐	5	Row6	7	83	Smartphone	4	2023-05-23

Output table (Table)

Rows: 208 | Columns: 8

☐	#	RowID	OrderID Number (integer)	CustomerID Number (integer)	Product String	Quantity Number (integer)	OrderDate Local Date	Year Number (integer)	Month (Number) Number (integer)	Day of Week (Number) Number (integer)
☐	1	Row2	3	15	Tablet	1	2025-01-01	2025	1	4
☐	2	Row3	4	72	Sma...	1	2023-10-09	2023	10	2
☐	3	Row4	5	61	Sma...	3	2023-09-08	2023	9	6
☐	4	Row5	6	21	Head...	2	2023-08-17	2023	8	5
☐	5	Row6	7	83	Sma...	4	2023-05-23	2023	5	3

Date&Time Part Extractor

เลือก Column ในการ Extract Date&Time →

เลือก Field (รูปแบบ) ที่จะ Extract ข้อมูล →

Date&Time Part Extractor



Dialog - 4:12 - Date&Time Part Extractor

Column selection

Date&Time column
OrderDate

Date and time fields

Field	Column name	
Year	Year	↑ ↓ 🗑️
Month (Number)	Month (Number)	↑ ↓ 🗑️
Day of Week (Number)	Day of Week (Number)	↑ ↓ 🗑️

+ Add field

Localization (month and day names, etc.)

Locale
English (United States)

Cancel Ok and Execute

Year

Year

Year (Week-based)

Quarter

Month (Number)

Month (Name)

Week

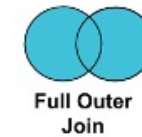
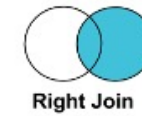
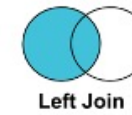
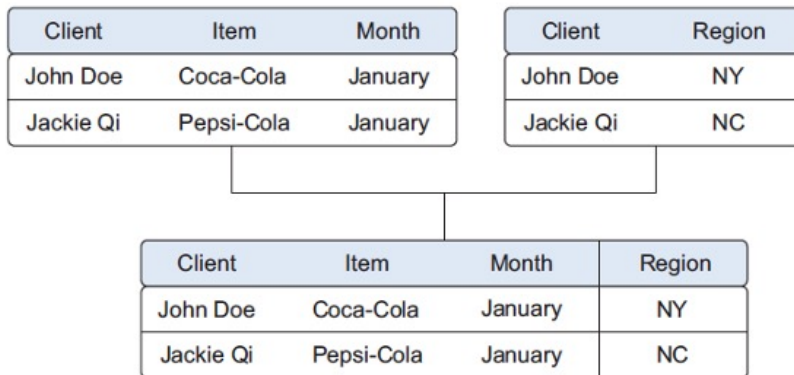
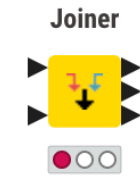
Day of Year

Day of week [1-7]

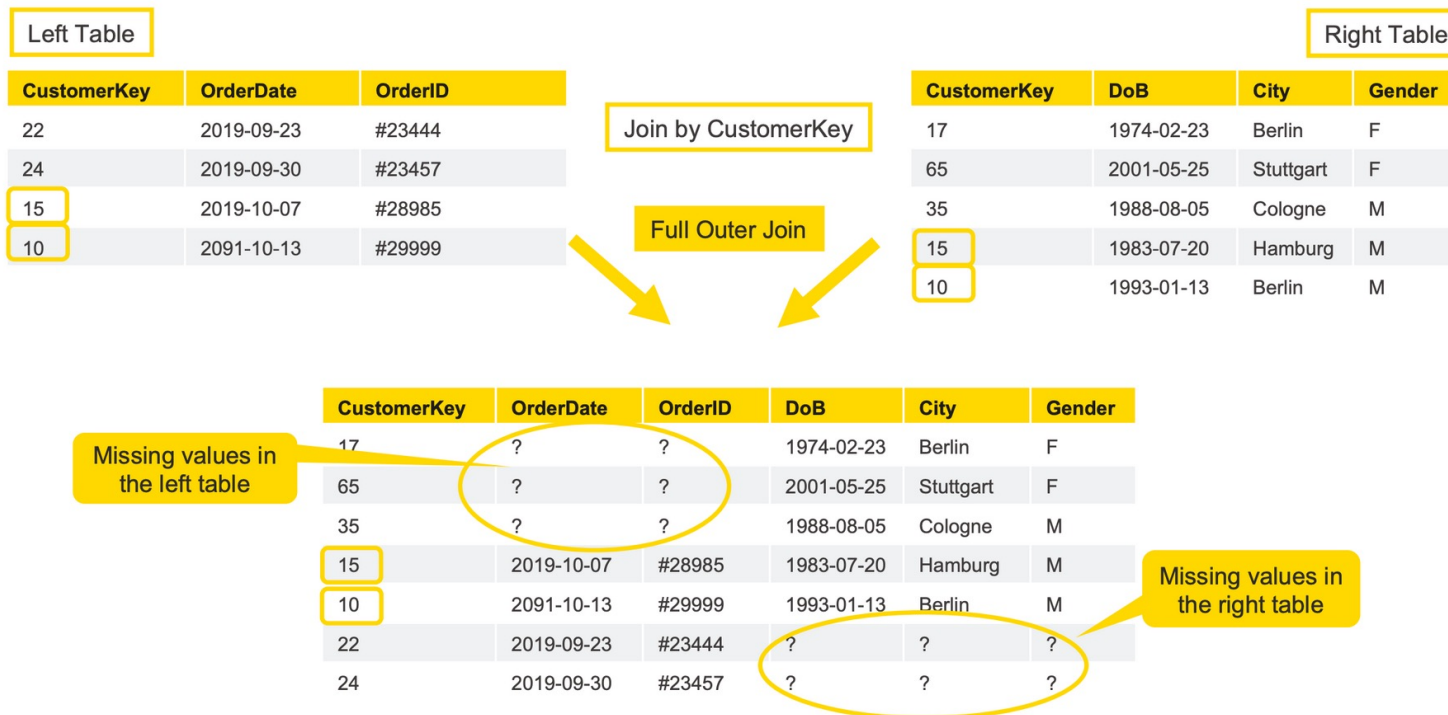


OrderDate Local Date	Year Number (inte...	Month Number (inte...	Date Number (inte...	Year (#1) Number (inte...	Month (N... Number (inte...	Day of W... Number (inte...
2025-01-01	2025	1	1	2025	1	4
2023-10-09	2023	10	9	2023	10	2
2023-09-08	2023	9	8	2023	9	6
2023-08-17	2023	8	17	2023	8	5
2023-05-23	2023	5	23	2023	5	3
2023-04-15	2023	4	15	2023	4	7
2024-11-23	2024	11	23	2024	11	7
2024-07-12	2024	7	12	2024	7	6
2023-03-31	2023	3	31	2023	3	6
2024-08-27	2024	8	27	2024	8	3
2024-03-08	2024	3	8	2024	3	6
2023-02-02	2023	2	2	2023	2	5
2023-01-31	2023	1	31	2023	1	3
2023-04-06	2023	4	6	2023	4	5

Joiner



Joiner



Joiner

Left Table

Filtered/Labeled Data (Table)										
Rows: 198		Columns: 8								
☐	#	RowID	OrderID Number (integ...	Custo... Number (integ...	Product String	Quantity Number (integ...	OrderDate String	Year Number (integ...	Month Number (integ...	Date Number (integ...
☐	111	Row...	113	1	Smartwatch	2	2024-05-26	2024	5	26
☐	145	Row...	147	1	Laptop	2	2024-06-25	2024	6	25
☐	191	Row...	193	1	Tablet	2	2023-03-29	2023	3	29
☐	15	Row...	17	2	Smartphone	1	2023-08-12	2023	8	12
☐	19	Row...	21	2	Laptop	4	2023-01-28	2023	1	28
☐	57	Row...	59	2	Headphones	3	2024-01-06	2024	1	6
☐	71	Row...	73	2	Laptop	4	2024-04-09	2024	4	9

Right Table

Input with replaced strings (Table)							
Rows: 200		Columns: 5					
☐	#	RowID	CustomerID Number (integer)	Name String	Email String	Phone String	City String
☐	1	Row0	1	Suwannee	suwannee1@hotmail.com	0836666371	Udon Thani
☐	2	Row1	2	Chaiwat	chaiwat2@gmail.com	0851180162	Nakhon Ratchasima

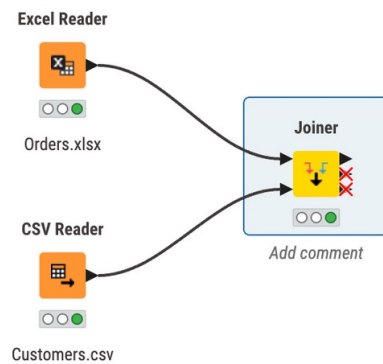
Join Table

Join result (Table)

Rows: 198 | Columns: 9

☐	#	RowID	OrderID Number (inte...)	Customer... Number (inte...)	Product String	Quantity Number (inte...)	OrderDate String	Name String	Email String	Phone String	City String	☐
☐	1	Row...	113	1	Smartwatch	2	2024-05-26	Suwannee	suwannee1@...	0836666371	Udon Thani	
☐	2	Row...	147	1	Laptop	2	2024-06-25	Suwannee	suwannee1@...	0836666371	Udon Thani	
☐	3	Row...	193	1	Tablet	2	2023-03-29	Suwannee	suwannee1@...	0836666371	Udon Thani	
☐	4	Row...	17	2	Smartphone	1	2023-08-12	Chaiwat	chaiwat2@g...	0851180162	Nakhon Ratc	
☐	5	Row...	21	2	Laptop	4	2023-01-28	Chaiwat	chaiwat2@g...	0851180162	Nakhon Ratc	
☐	6	Row...	59	2	Headphones	3	2024-01-06	Chaiwat	chaiwat2@g...	0851180162	Nakhon Ratc	
☐	7	Row...	73	2	Laptop	4	2024-04-09	Chaiwat	chaiwat2@g...	0851180162	Nakhon Ratc	

Joiner



Dialog - 4:8 - Joiner (join customer and order)

Matching Criteria

Match
All of the following Any of the following

Top input ('left' table)	Bottom input ('right' table)
CustomerID	CustomerID

+ Add matching criterion

Compare values in join columns by
Value and type

Include in Output

☒ Matching rows
☐ Left unmatched rows
☐ Right unmatched rows

Output Columns

Top input ('left' table)
Manual Wildcard Regex

Cancel Ok and Execute

Dialog - 4:8 - Joiner (join customer and order)

Output Columns

Top input ('left' table)
Manual Wildcard Regex

Search Aa

Excludes	Includes
Year Month Date	OrderID CustomerID Product Quantity OrderDate

Bottom input ('right' table)
Manual Wildcard Regex

Search Aa

Excludes	Includes
CustomerID	Name Email Phone City

☐ Merge join columns

Cancel Ok and Execute

Scroll
Down

Sorter

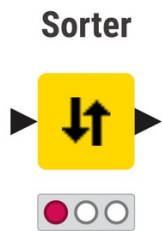
Sort by CustomerID Ascending

Sorted Table (Table)

Rows: 198 | Columns: 9

<

Sorter



Sort By

Order by

Dialog - 4:19 - Sorter

Sorting

Criterion 1 ↑ ↓ ✕

Column
CustomerID ▼

Order
Ascending Descending

⊕ Add sorting criterion

[Show advanced settings](#)

Cancel Ok and Execute

Aggregate

Row Aggregator

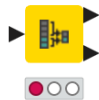


Diagram illustrating a data flow process:

- Excel Reader (Orders.xlsx) and CSV Reader (Customers.csv) feed into a Joiner.
- The Joiner output feeds into a Sorter.
- The Sorter output feeds into a Row Aggregator.
- The Row Aggregator has a comment "Add comment".

Dialog - 5:9 - Row Aggregator

Category column: City

Aggregation: ☐ Occurrence count ☒ Sum ☐ Average ☐ Minimum ☐ Maximum

Aggregation columns: Manual Wildcard Regex Type

Search: Aa

Excludes: OrderID, CustomerID

Includes: Quantity

Weight column: No value selected

☐ Additional "grand totals" at second output port

[Show advanced settings](#)

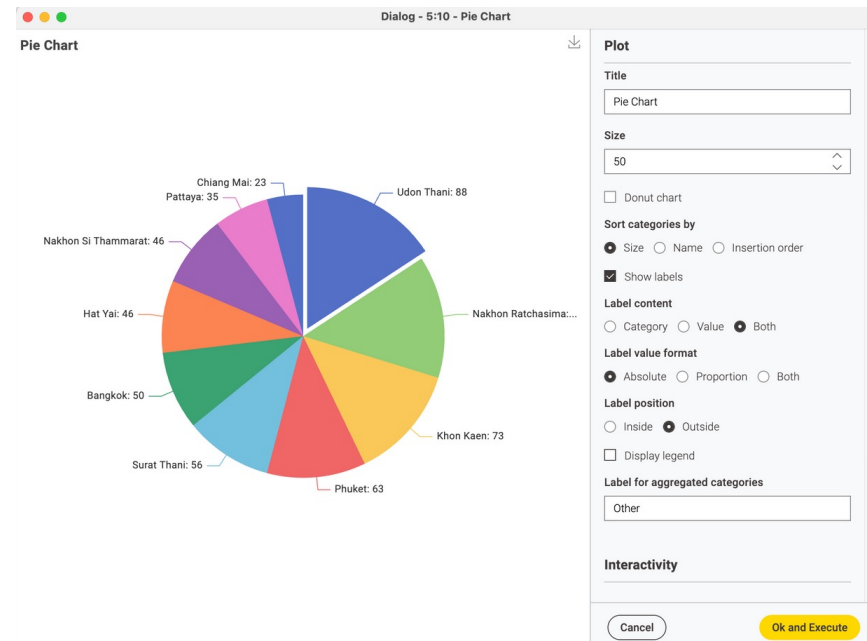
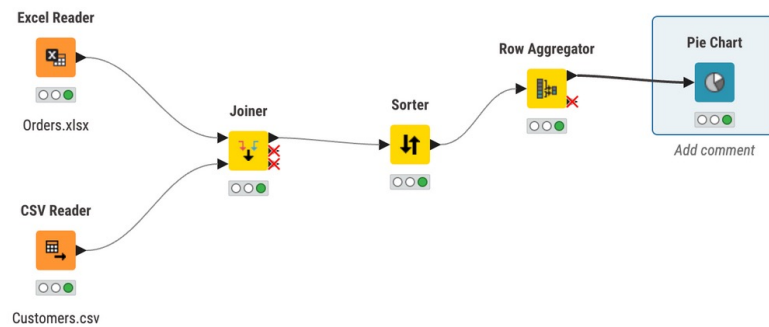
Cancel Ok

► 1: Aggregation results ✖ 2: Grand-total results Flow Variables

Rows: 10 | Columns: 2

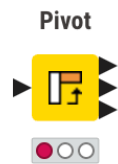
☒	#	RowID	City String	Quantity Number (integer)
☐	1	Row0	Bangkok	50
☐	2	Row1	Chiang Mai	23
☐	3	Row2	Hat Yai	46
☐	4	Row3	Khon Kaen	73
☐	5	Row4	Nakhon Ratchasima	78
☐	6	Row5	Nakhon Si Thammarat	46
☐	7	Row6	Pattaya	35
☐	8	Row7	Phuket	63
☐	9	Row8	Surat Thani	56
☒	10	Row9	Udon Thani	88

Basic Plot



Pivot

ตัวอย่าง

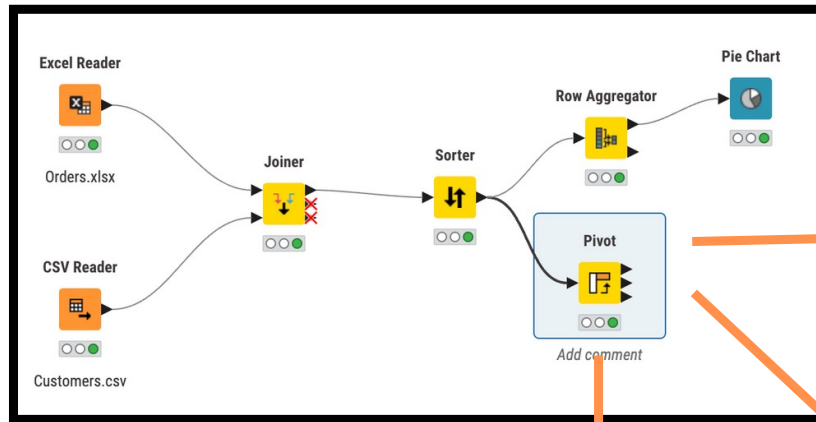


OrderNr	ProductNr	StoreType	Quantity
23893756	I-163-2017	Onsite Store	2
23893756	B-172-2005	Onsite Store	3
23893756	W-181-2003	Onsite Store	1
23893762	I-163-2017	Online Store	8
...			
23893787	F-160-1993	Online Store	3
23893787	I-163-2017	Online Store	4



ProductNr	Online Store	Onsite Store
I-163-2017	23	56
B-172-2005	65	0
...		
W-181-2003	33	2
V-031-1992	72	44

Sum of Quantity



Groups Pivots Manual Aggregation

Pivot settings

Available column(s)

Filter

- OrderID
- CustomerID
- Quantity
- OrderDate
- Name
- Email
- Phone
- City

Pivot column(s)

Filter

- Product

☒ Ignore missing values ☐ Append overall totals ☒ Ignore domain

Dialog - 5:12 - Pivot

Settings Description Flow Variables Job Manager Selection Memory Policy

Groups Pivots Manual Aggregation

Group settings

Available column(s)

Filter

- OrderID
- Product
- Quantity
- OrderDate
- Name
- Email
- Phone
- City

Group column(s)

Filter

- CustomerID

Advanced settings

Column name: Pivot name Aggregation name: Aggregation method (column name) ☐ Sort

Maximum unique values per group 10,000 Value delimiter ☐ Process in memory ☐ Retain row order ☐ Enable

OK Apply Cancel ?

Groups Pivots Manual Aggregation

Aggregation settings

Available columns

- OrderID
- Quantity
- OrderDate
- Name
- Email
- Phone
- City

Select

add >>

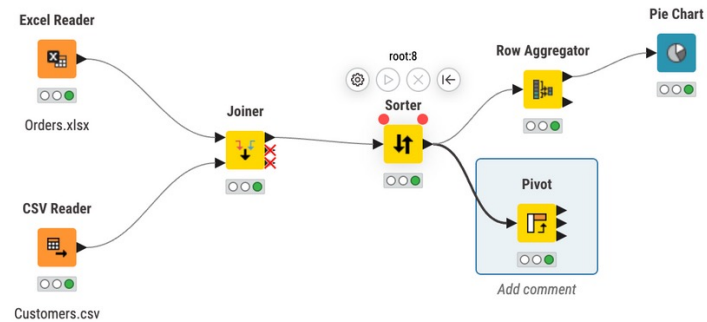
add all >>

<< remove

<< remove all

To change multiple columns use right mouse

Column	Aggregation (click to change)
Quantity	Mean

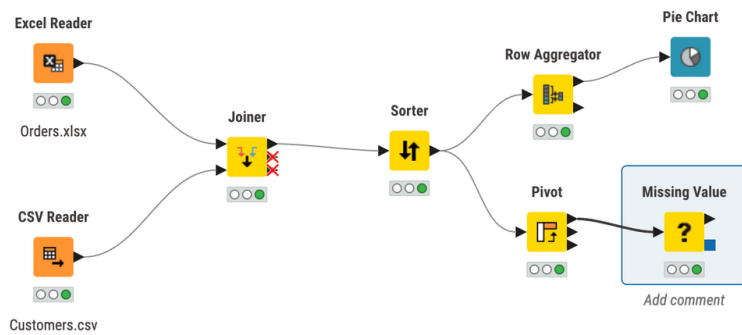


► 1: Pivot table ► 2: Group totals ► 3: Pivot totals ☒ Flow Variables

Rows: 87 | Columns: 6

Table ☐ Statistics ☐

☐	#	RowID	CustomerID Number (integer)	Laptop Number (integer)	Smartwatch Number (integer)	Tablet Number (integer)	Headphones Number (integer)	Smart Number
☐	1	Row0	1	2	2	2	?	?
☐	2	Row1	2	8	?	?	3	1
☐	3	Row2	3	?	4	4	?	4
☐	4	Row3	4	4	?	5	?	1
☐	5	Row4	5	4	?	?	4	?
☐	6	Row5	6	?	?	1	3	?
☐	7	Row6	7	?	?	?	4	4
☐	8	Row7	8	2	2	?	?	4
☐	9	Row8	9	8	?	?	4	?
☐	10	Row9	11	4	?	?	?	?
☐	11	Row10	12	2	?	?	?	?



Dialog - 5:14 - Missing Value

Default | Column Settings | Flow Variables

Number (integer)

Fix Value [dropdown]

Value [input field with '0']

OK - Execute | Apply | Cancel | ?

► 1: Output table ■ 2: PMML Transformations 📄 Flow Variables

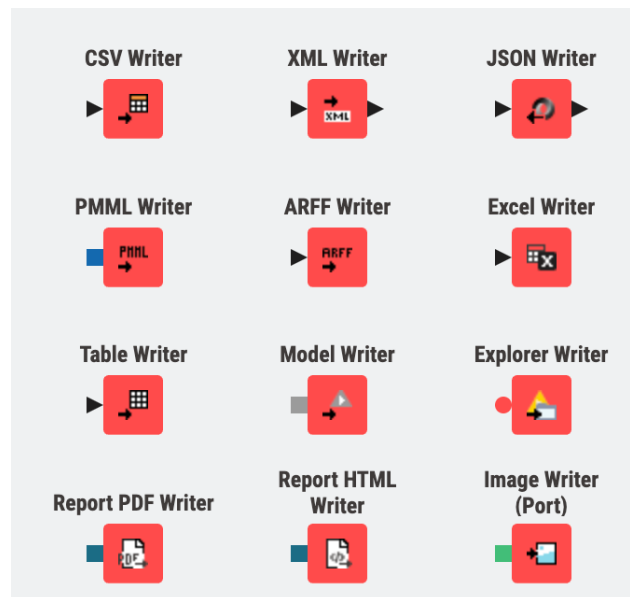
Rows: 87 | Columns: 6

Table | 📄 | Statistics | 📄

☐	#	RowID	CustomerID <i>Number (integer)</i>	Laptop <i>Number (integer)</i>	Smartwatch <i>Number (integer)</i>	Tablet <i>Number (integer)</i>	Headphones <i>Number (integer)</i>	Smartphone <i>Number (integer)</i>
☐	1	Row0	1	2	2	2	0	0
☐	2	Row1	2	8	0	0	3	1
☐	3	Row2	3	0	4	4	0	4
☐	4	Row3	4	4	0	5	0	1
☐	5	Row4	5	4	0	0	4	0
☐	6	Row5	6	0	0	1	3	0
☐	7	Row6	7	0	0	0	4	4
☐	8	Row7	8	2	2	0	0	4
☐	9	Row8	9	8	0	0	4	0
☐	10	Row9	11	4	0	0	0	0
☐	11	Row10	12	2	2	2	2	2

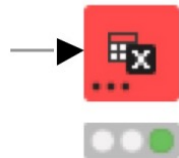
Exporting Data

- Exporting Data Node จะเป็น Node สีแดง
- ใช้ในการ นำข้อมูลที่เราทำการ Manipulate & Analysis



Excel Writer

Excel Writer



Dialog - 4:9 - Excel Writer (export to excel)

Settings Encryption Flow Variables Job Manager Selection

File format & output location

Excel format XLSX

Write to Local File System

File /Lab_Customer_Orders.xlsx **Browse...**

Write options ☐ Create missing folders If exists: ☒ overwrite ☐ append ☐ fail

There exists a file with the specified path
Orders.xlsx that will be overwritten.

Sheets

1. sheet name default_1

If sheet exists ☐ overwrite ☐ append ☐ fail

Names and IDs

☐ Write row key

☒ Write column headers

☒ Don't write column headers if sheet exists

Missing value handling

☐ Replace missing values by

Formulas

☐ Evaluate formulas (leave unchecked if uncertain; see node description for details)

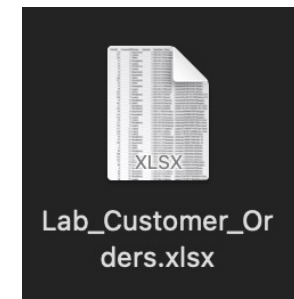
Layout

☐ Autosize columns

☒ Portrait ☐ Landscape A4 - 210x297 mm

☐ Open file after execution

OK - Execute Apply Cancel ?



Lab1: Customer Orders

- ไฟล์ Customers.csv และ Orders.xlsx ซึ่งจะประกอบด้วยข้อมูลลูกค้าและรายการที่สั่งซื้อ
- เตรียมข้อมูลเพื่อทราบว่าลูกค้าท่านใดสั่งซื้ออะไรบ้างในวันไหนจำนวนเท่าไร

Lab2: Customer Orders

ขั้นตอน

- นำเข้าข้อมูล CSV และ XLSX ในโปรแกรม KNIME
- Validate OrderDate Column ในไฟล์ Orders.xlsx
 - ex: 2024/01/01 -> 2024-01-01
- แปลงเบอร์โทรศัพท์จาก +66- ให้เป็น 0
 - ex: +66-9XXXXXX -> 09XXXXXXXX
- นำข้อมูล 2 Tables มา JOIN กัน
- สุดท้ายนำข้อมูลออกไปเป็น Excel

Lab3: Auto Data set

- Create a KNIME workflow where you read the provided autos data set.
- Exclude the columns “normalized” and “bore”.
- Filter the rows on the price column and keep only instances describing cars that are less than \$10,000.
- Write the file out to csv and report back the number of rows.

Q&A